

GREEN to BROWN

Summer Vegetation Change of ECSD Parcel



2005



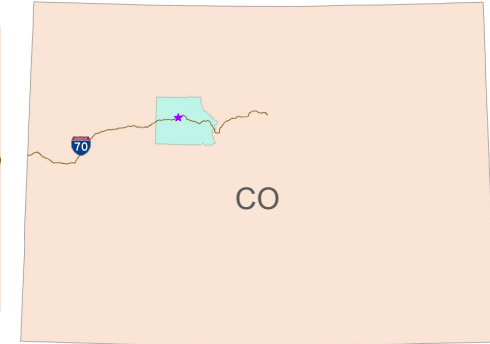
2020

By: Cassadie Mefferd

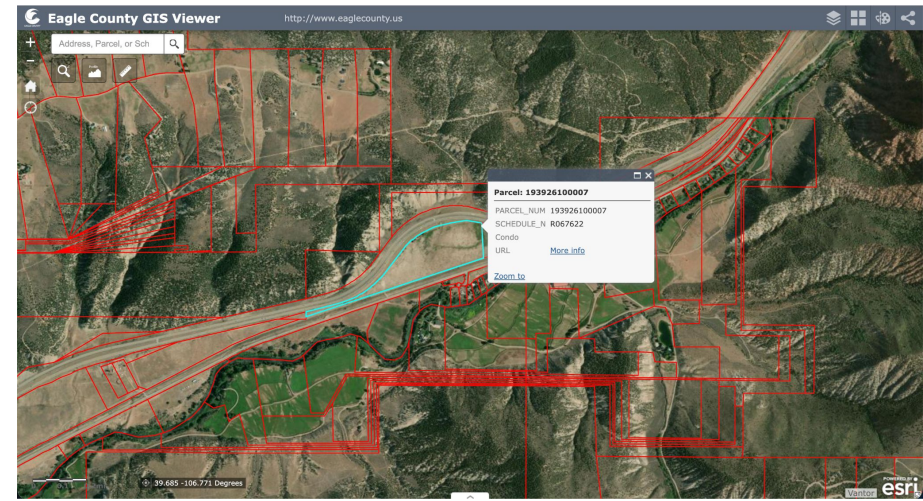
Study Objective

- Compare vegetation condition over time (2005), (2015), & (2025).
- Identify whether the site shows potential for recovery and highlight areas of opportunity
- Provide spatial evidence to support restoration planning and future decisions

- Eagle County School District - Harvest Innovation Grant
- Stakeholder Feasibility Meetings for 'Farm-to-School' Program.



Parcel #193926100007







9/23/2011

70

70

70

70

Eagle River

Eagle River

Warren Ditch

Eagle River

Hollingsworth Ditch

Eagle River

on Ditch





6/24/2017



Eagle River

Eagle River

Warren Ditch

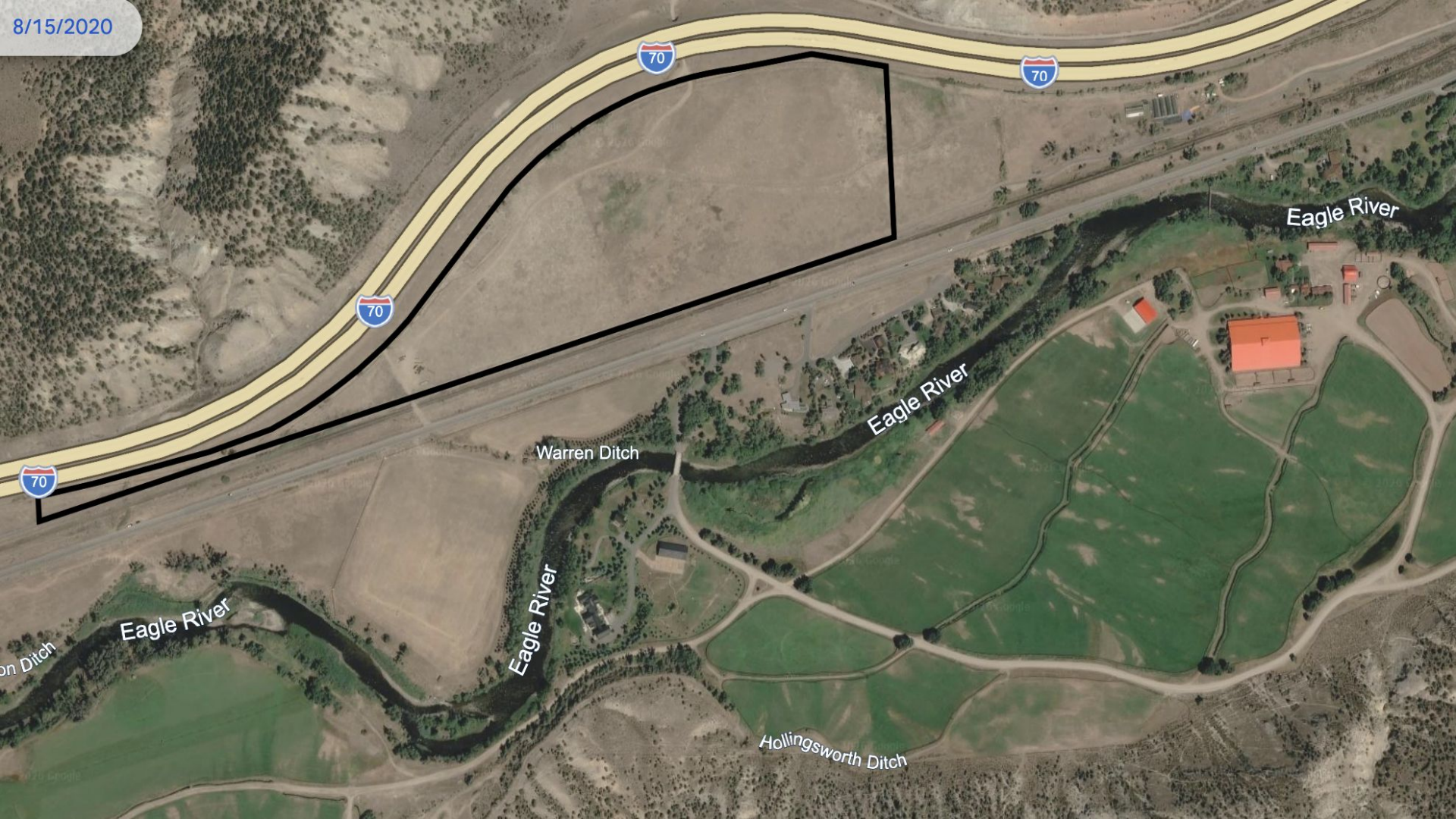
Eagle River

Hollingsworth Ditch

Eagle River

on Ditch





8/15/2020

70

70

70

70

Eagle River

Eagle River

Warren Ditch

Eagle River

Eagle River

on Ditch

Hollingsworth Ditch



9/22/2021



Eagle River

Eagle River

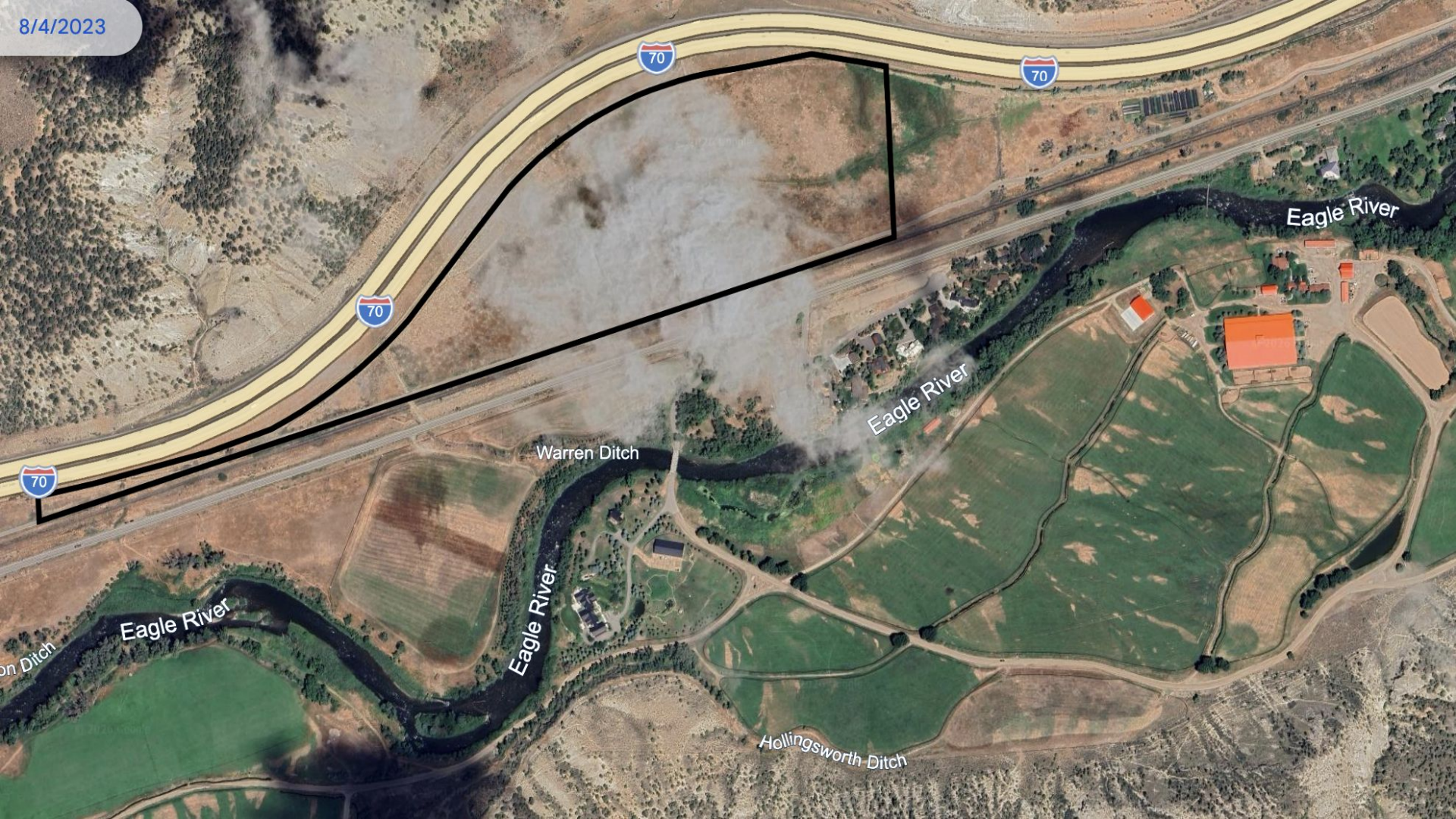
Warren Ditch

Eagle River

Hollingsworth Ditch

on Ditch

Eagle River



Data Sources

Study Boundary - **Polygon** - Shapefile

Eagle County Parcels - ArcGIS Online “Amy Keeley” (2019)

I used Landsat imagery in Google Earth Engine and created summer *composites* from *June 1st through August 31st* for each year.

I used **Landsat 5** for 2005 and **Landsat 8** for 2015 and 2025.

USGS Landsat 5 Level 2, Collection 2, Tier 1

USGS Landsat 8 Level 2, Collection 2, Tier 1

Analytical Methods

ArcGIS Pro

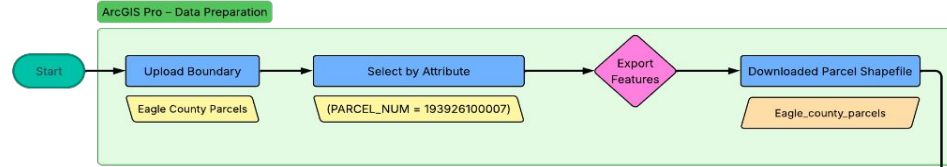
- Isolate selected parcel to create study boundary

Google Earth Engine

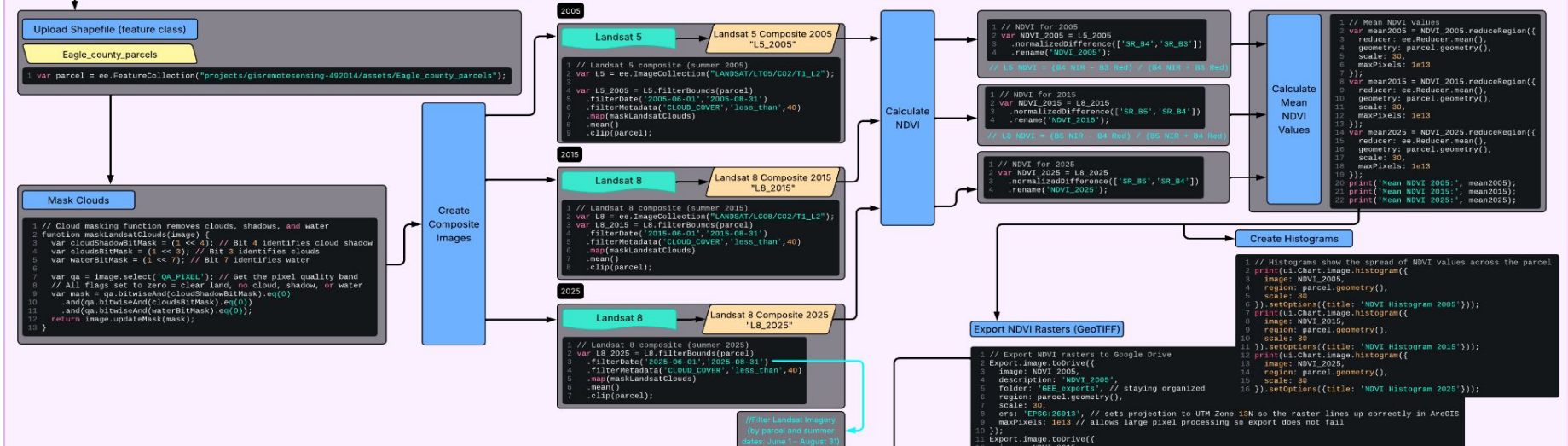
- Landsat imagery
- Create composites
- Calculated NDVI
- Build histograms

Analysis

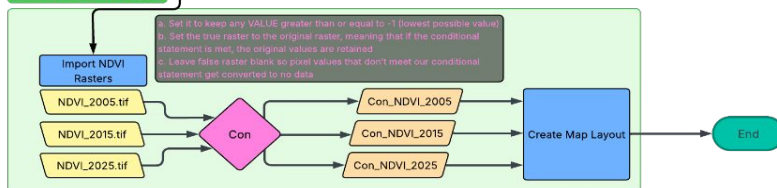
- Compared mean NDVI vegetation across 2005, 2015, and 2025



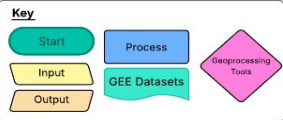
Google Earth Engine – Data Acquisition and NDVI Processing



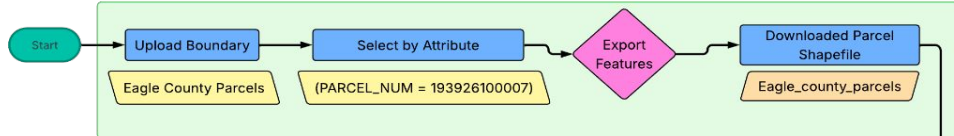
ArcGIS Pro – Final Mapping



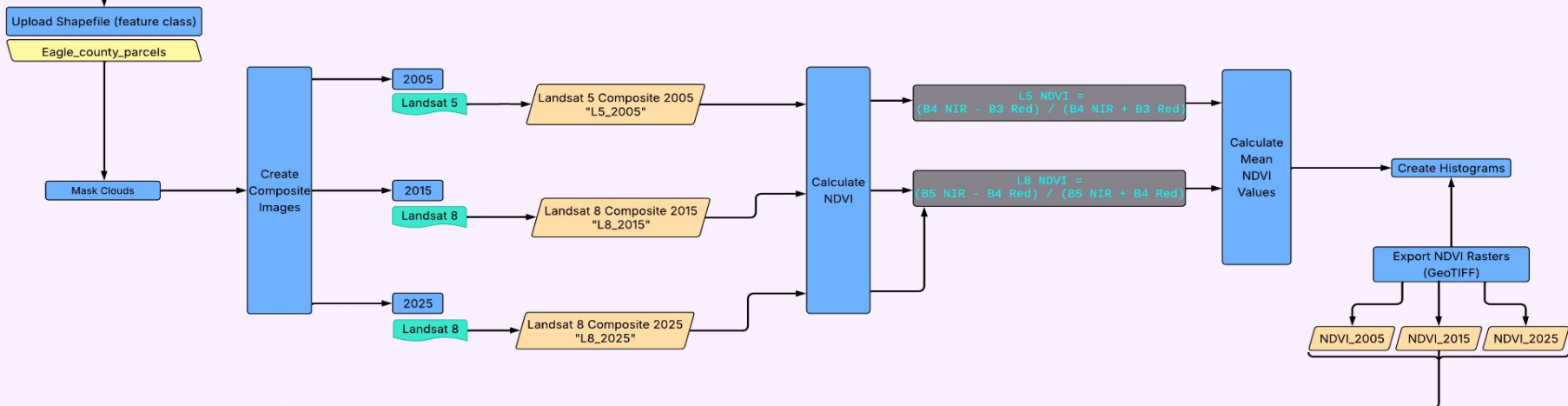
NDVI GEE Workflow



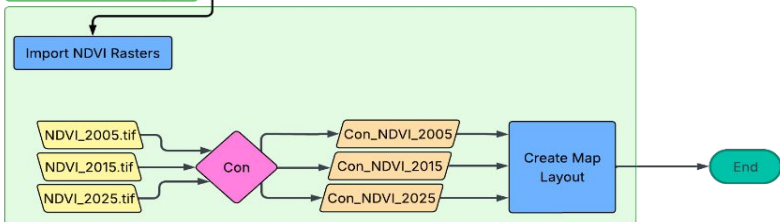
ArcGIS Pro – Data Preparation



Google Earth Engine – Data Acquisition and NDVI Processing



ArcGIS Pro – Final Mapping



Results

NDVI values across the parcel ranged from approximately 0.08 to approximately 0.30 across all three years, indicating generally sparse to moderate vegetation.

Mean NDVI values showed a **decline** over time:

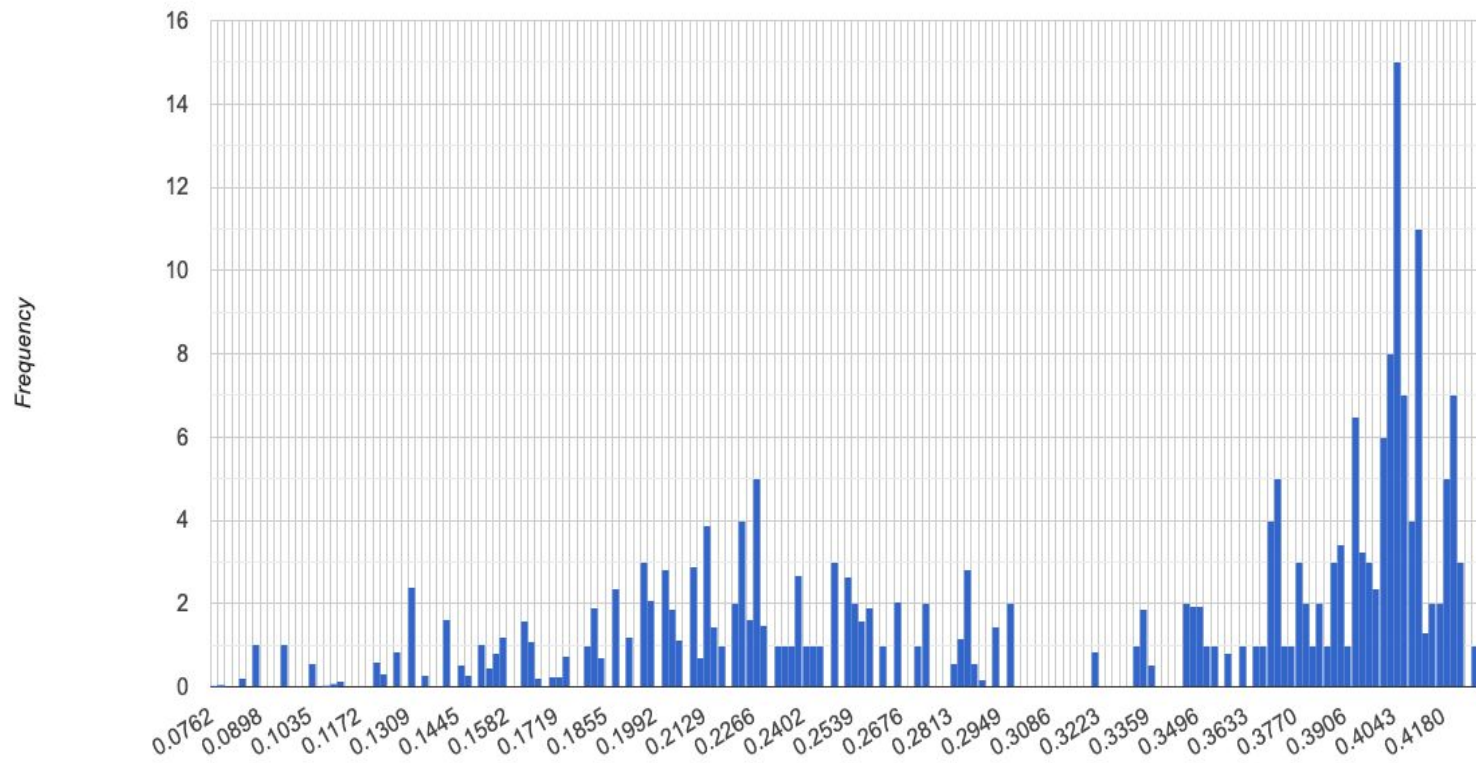
```
Mean NDVI 2005: JSON
▼Object (1 property) JSON
  NDVI_2005: 0.3196529857259515

Mean NDVI 2015: JSON
▼Object (1 property) JSON
  NDVI_2015: 0.2261265107619816

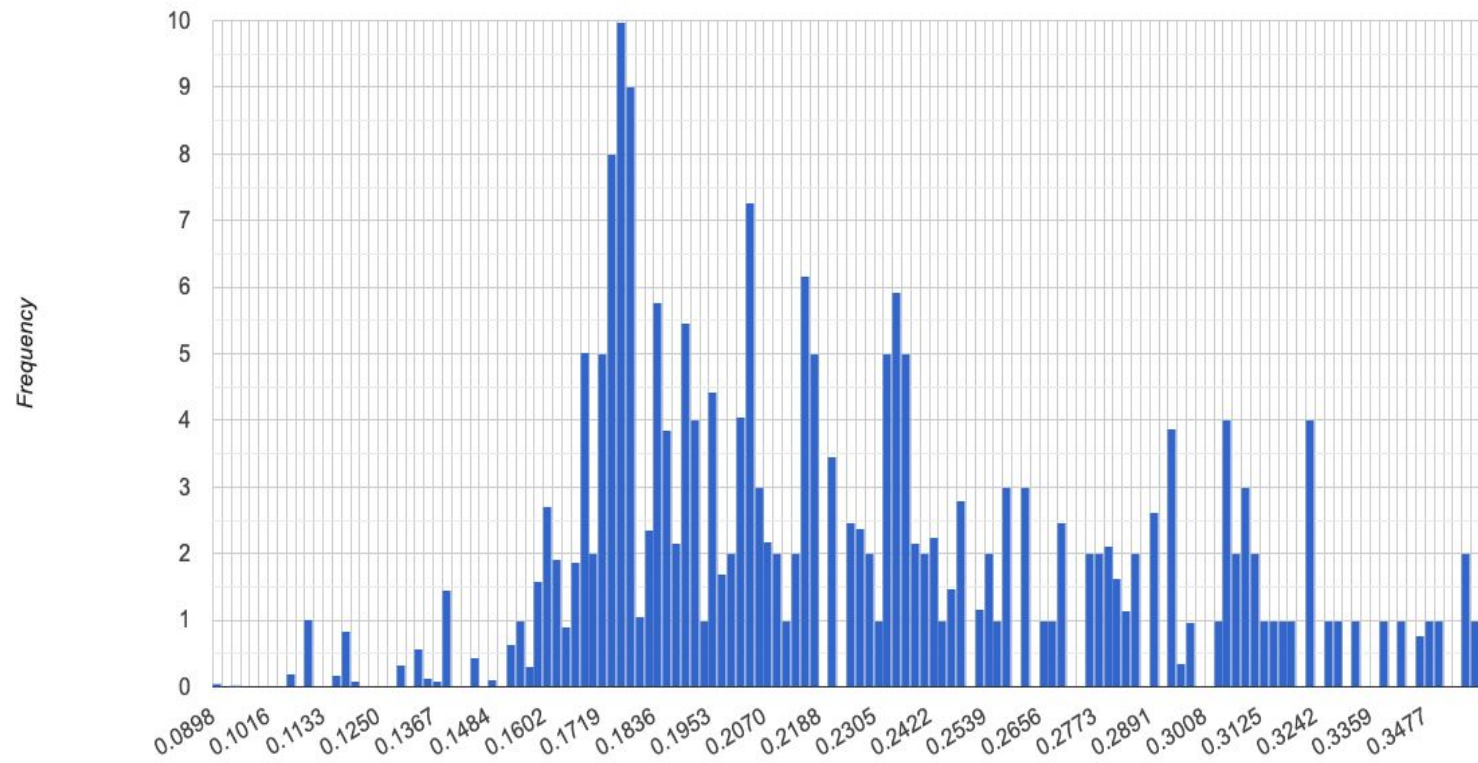
Mean NDVI 2025: JSON
▼Object (1 property) JSON
  NDVI_2025: 0.13675767482896273
```

- 2005 → 2015: NDVI ↓ (0.09); 29% decrease.
- 2015 → 2025: NDVI ↓ (0.09); 40% decrease.
- Overall, that's a 57% decline in vegetation over 20 years.

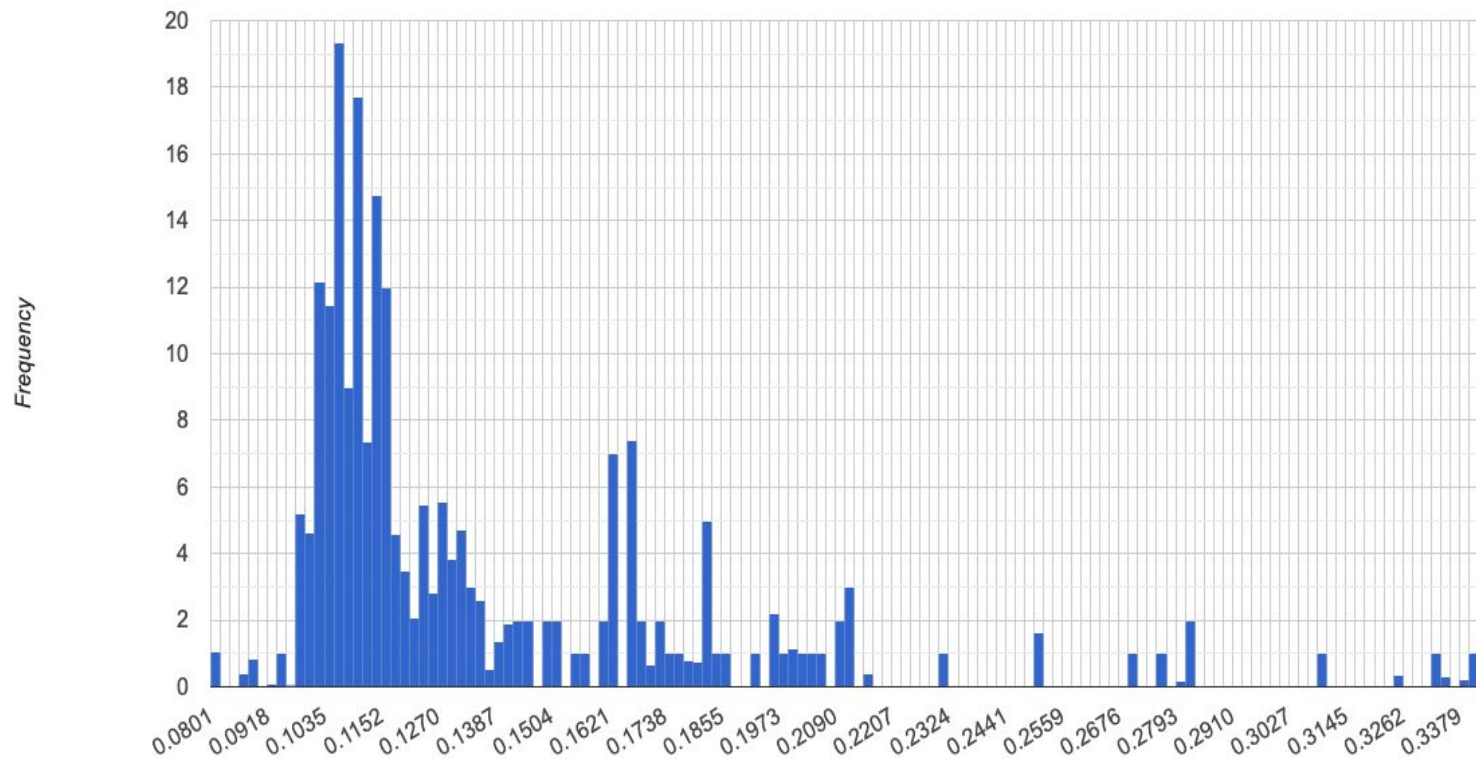
NDVI Histogram 2005



NDVI Histogram 2015

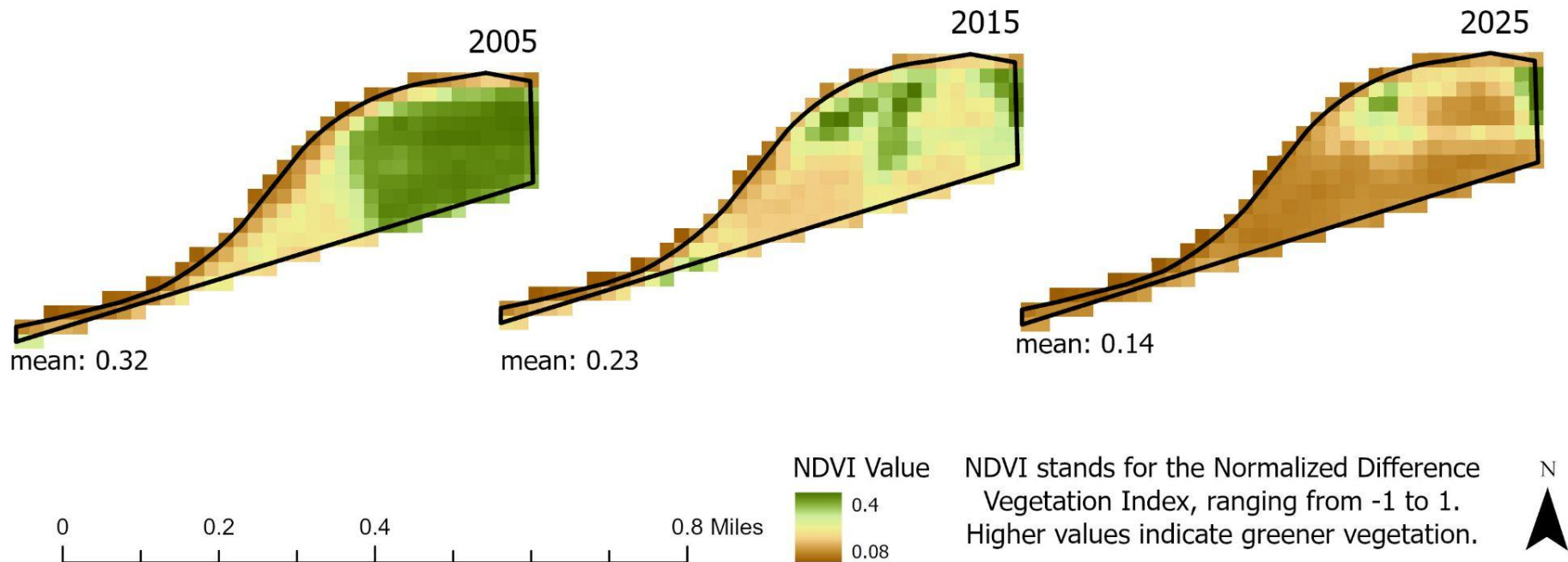


NDVI Histogram 2025



From Green to Brown: Summer Vegetation Change of ECSD Parcel Over the Past 20 years

Each image represents a summer composite (June 1st - August 31) for the given year.



Discussion

Limitations

- Does not capture soil or moisture
- Moderate resolution (30 m Landsat pixels)
- Minor variation between GEE and ArcGIS calculations

Recommendations

- Begin vegetation regeneration and improved land management
 - Focus on areas with higher NDVI as starting points
- Consider partnerships with local ranchers
- Strong potential for education due to location
- Expand analysis with NDMI or field data

